Support Coupling in Combined Length-Debiased and Positive-Preserving Preference Optimization

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The Problem: Confident, Not Correct

DPO [5] improves judge-rated quality but also reshapes **how confidently** a model speaks — independent of correctness. A wrong answer that loses its hedging becomes an **overconfident error** (confidently wrong), the worst failure in health/legal settings: it removes the cue a reader would use to seek a second opinion.

Three Questions

1. Does vanilla DPO **suppress uncertainty** and worsen calibration on incorrect answers?
2. Do **SamPO** [3] (length debiasing) and **DPOP** [4] (positive preservation) mitigate it?
3. When combined, do they **compose**? If not, why?

Reproduce → Critique → Extend

- **Reproduce:** DPO suppresses uncertainty; SamPO and DPOP each mitigate it.
- **Critique:** their *naive* combination is **sub-additive**.
- **Extend:** we diagnose *support coupling* and propose a **decoupled-support** fix.

Overview

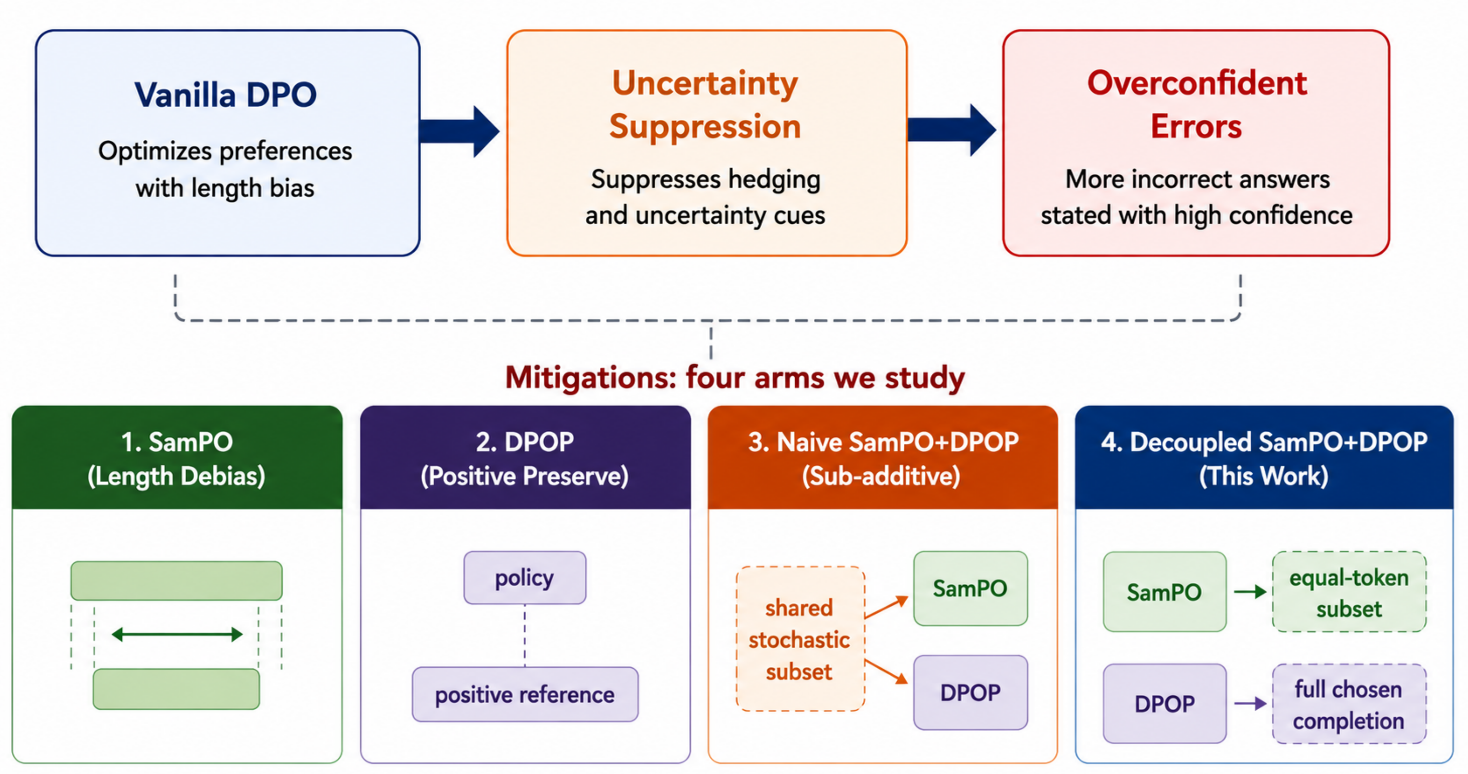

Figure 1. Vanilla DPO → uncertainty suppression → overconfident errors, and the four mitigation arms we study.

References

[1] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021.

[2] Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods, 2022.

[3] Junru Lu, Jiazheng Li, Siyu An, Meng Zhao, Yulan He, Di Yin, and Xing Sun. Eliminating biased length reliance of direct preference optimization via down-sampled kl divergence, 2024.

[4] Arka Pal, Deep Karkhanis, Samuel Dooley, Manley Roberts, Siddartha Naidu, and Colin White. Smaug: Fixing failure modes of preference optimisation with dpo-positive, 2024.

[5] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model, 2024.

Method: Two Building Blocks

SamPO compares **equal-token subsets** S (Δ_S, length-debiased); DPOP **anchors** the chosen completion above the reference ($P = \max(0, \log \frac{\pi_{\text{ref}}(y_w)}{\pi_\theta(y_w)})$). The six arms differ only in the loss:

Arm	Comparison	Preservation
baseline	—	—
vanilla	Δ_{full}	—
SamPO	Δ_S	—
DPOP	Δ_{full}	P_{full}
naive	Δ_S	P_{subset}
decoupled	Δ_S	$\mathbf{P}_{\text{full}}$

Diagnosis: Support Coupling & the Fix

The naive composition evaluates DPOP's anchor on the *same* stochastic subset S SamPO resamples each step, so each token is preserved with probability $|S_w|/|C|$ — only a **random fragment** is anchored and the hedging tokens are diluted. *No λ fixes a support mismatch.* The fix keeps SamPO's subset for the comparison but restores DPOP's anchor to the *full* completion:

$$L_{\text{decoupled}} = -\log \sigma\big(\beta[\,\Delta_S - \lambda P_{\text{full}}\,]\big).$$

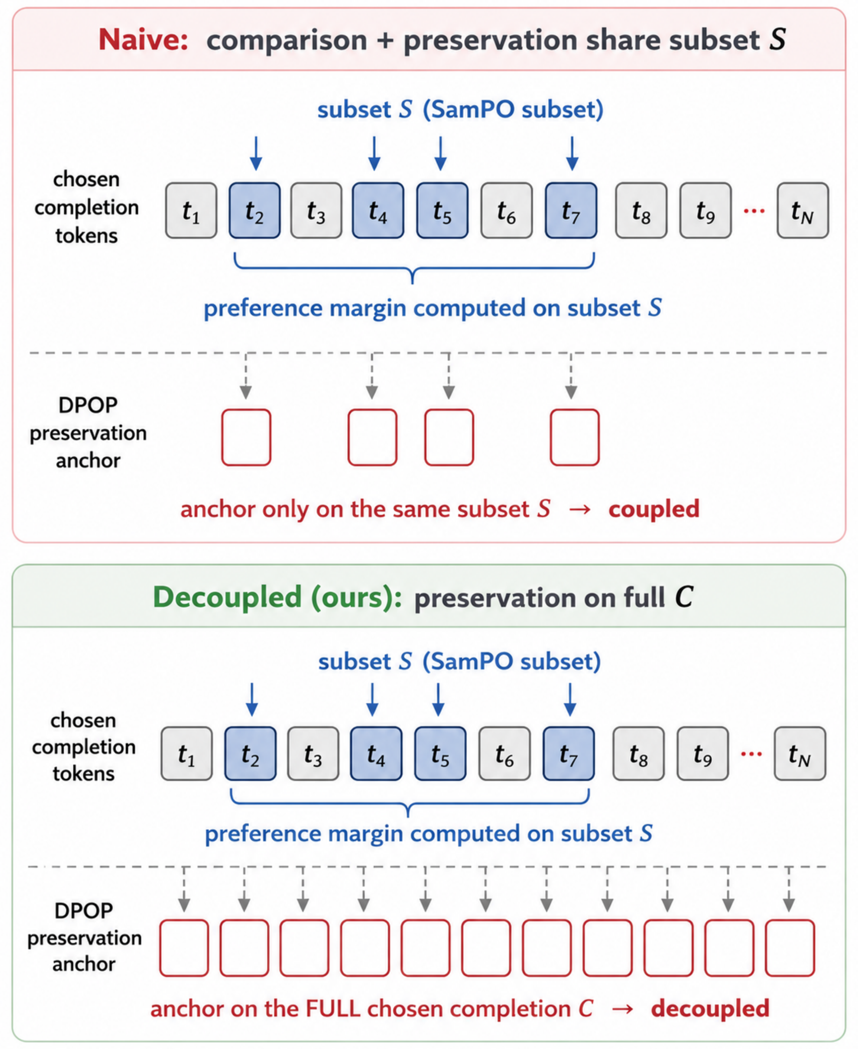

Figure 2. Naive: comparison and preservation share subset S. Decoupled: preservation moves to the full completion C.

Limitations

- 3 seeds, single dataset (TruthfulQA); the ~200-token cap makes $|S_w| \approx |C|$ and compresses SamPO.
- AI-assisted factuality / calibration labels, so these rates are exploratory rather than human-verified.

Results: Uncertainty & Calibration

Qwen3-0.6B + LoRA [1], judged by Qwen2.5-7B-Instruct on TruthfulQA [2] (80/80 split, 3 seeds).

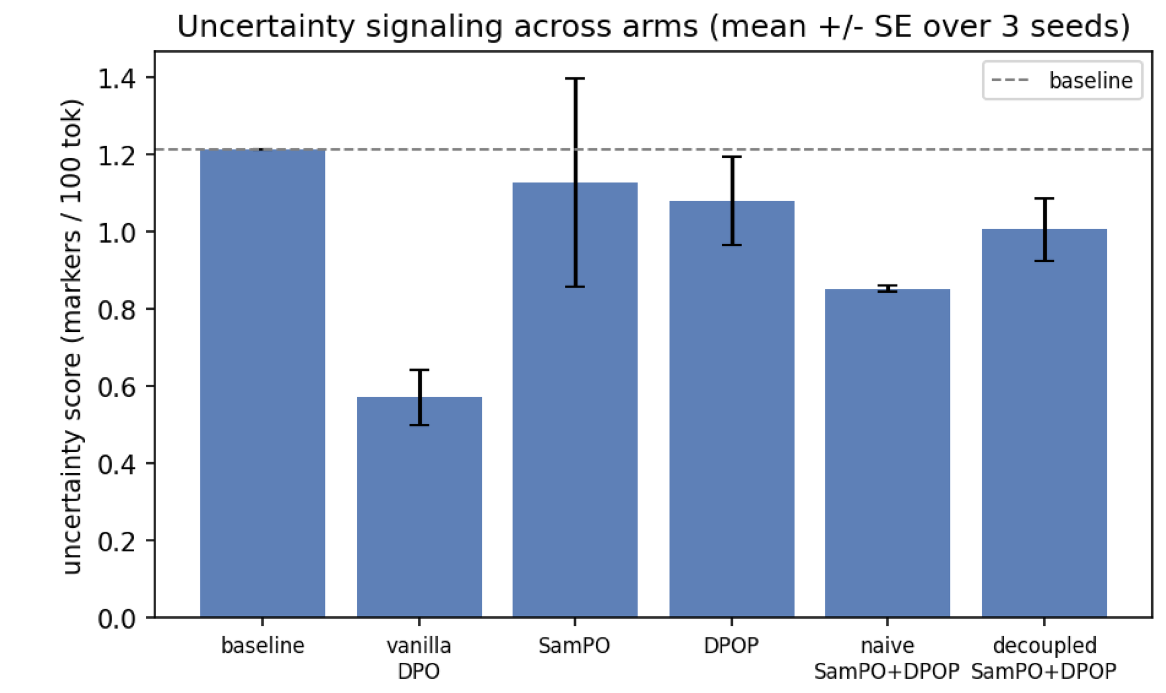

Figure 3. Vanilla DPO collapses uncertainty ($1.21 \to 0.57$, -53%); mitigations restore it, the naive combo least; decoupled recovers most (1.00).

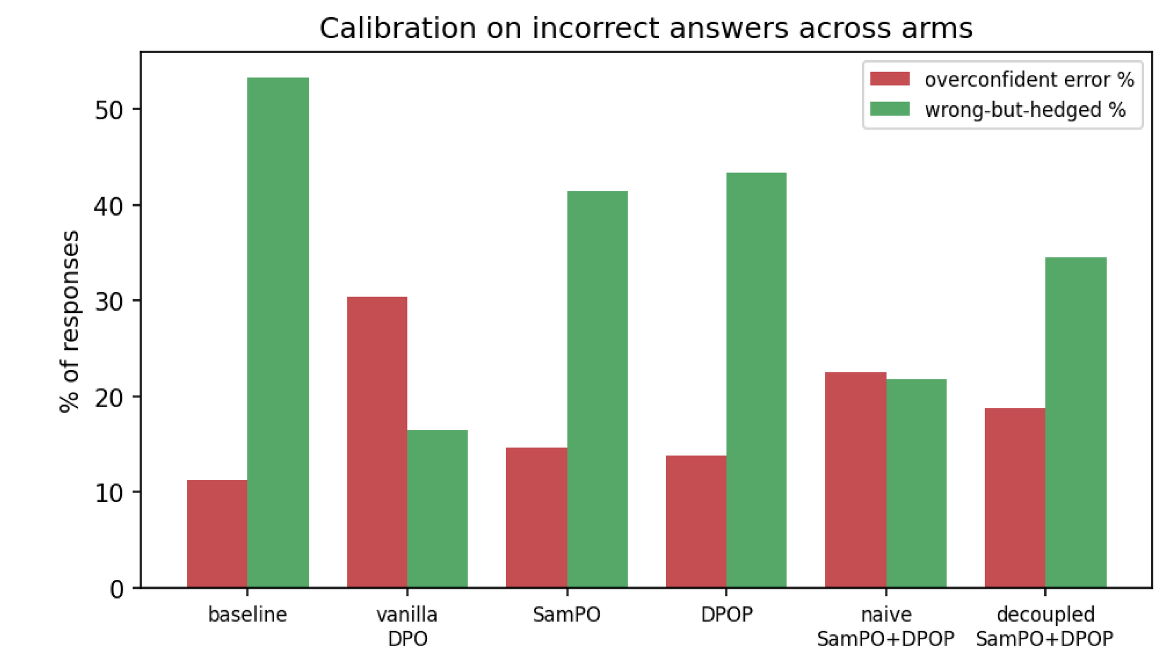

Figure 4. Overconfident errors on incorrect answers: vanilla nearly triples them ($11.2\% \to 30.4\%$); decoupling shifts the naive combo back toward hedged errors ($22.5\% \to 18.8\%$, $p = 0.028$ paired).

Application

A medical red-team triage on TruthfulQA `Health` prompts reuses the overconfidence signal: vanilla DPO yields the most overconfident health answers, DPOP the fewest — qualitative triage, not a clinical claim.

Takeaway

- DPO changes **presentation, not correctness**: factuality is flat across all arms.
- Naive SamPO+DPOP is sub-additive — a **support** problem, not a λ problem.
- Decoupling repairs the naive combo but **does not beat** standalone SamPO/DPOP and costs a little win-rate: a diagnosis + minimal fix, not a new SOTA method.